

ROUTE2ZERO

AI/ML-assisted planning for a just e-jeepney transition.

Which Metro Manila corridors should be validated and prioritized first — and how robust is that decision?

1,522

route-direction records screened

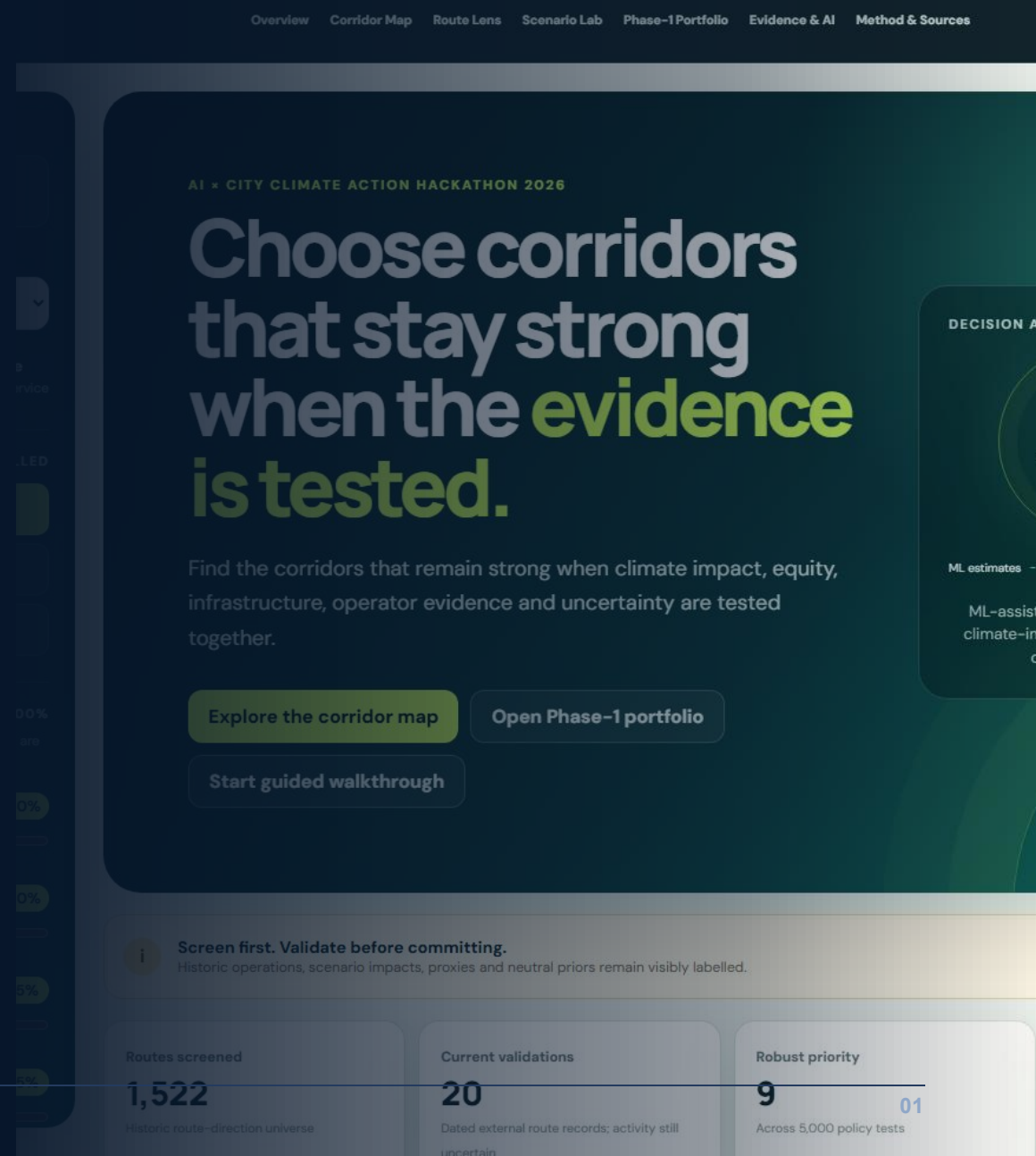
9

robust-priority corridors

20

dated OSM validations

Team Larpers · Metro Manila · AI x City Climate Action Hackathon 2026



WHY THIS MATTERS

Transport electrification is a climate opportunity — prioritization is the bottleneck.

34%

of Philippine GHG emissions came from transport in 2015

80%

of transport emissions were attributed to road transport

NATIONAL DIRECTION

50% EV adoption by 2040

A roadmap target — not an achieved value and not route-level proof.

Cities still must decide where limited validation, planning, charging, and transition support create the strongest public value first.

2025 DOE direction: distribution utilities integrate EV charging demand into development planning.

THE PLANNING GAP

The evidence for one corridor decision lives in different systems.

Route + service

historic schedules, current status

Climate + energy

vehicle, grid, and operating assumptions

Equity + access

population and validated area evidence

Charging

mapped proximity versus verified capacity

Operators

fleet, depot, finance, governance

Uncertainty

data quality and policy sensitivity

Evidence does not agree by default.

High activity can still mean weak charging evidence.

High equity can rest on outdated service records.

A top score can move when policy priorities change.

Route2Zero integrates the evidence — it does not erase disagreement.

DECISION FRAME

Route2Zero answers one city decision.

Which corridors should Metro Manila validate and prioritize for e-jEEPney transition — and how robust is that recommendation?

- Where is the climate value?
- Who benefits — and what evidence supports that?
- Can charging and operators support the transition?
- Does the recommendation survive changing assumptions?

Users: LGUs · DOTr/LTFRB · operators · energy partners · finance · riders and communities

Output: priority · evidence · climate range · stability · portfolio status · next validation action

PRODUCT PROMISE

From a leaderboard to a city electrification decision laboratory.

01	02	03	04	05	06
Screen	Estimate	Quantify	Test	Stress-test	Assemble
1,522 route directions	one missing service input	climate + energy scenarios	equity, charging, operators	rank stability + evidence	constrained Phase-1 portfolio

Planning & Evidence Assistant

Explains precomputed evidence, compares scenarios, and turns missing evidence into a fieldwork queue.

Humans decide

Policy weights, constraints, evidence acceptance, and implementation remain explicit city choices.

AI has analytical work — policy authority stays human.

MACHINE LEARNING

Service-intensity estimate
Anomaly flag
Corridor typology

DETERMINISTIC MODELS

Climate + energy
Evidence confidence
Sensitivity + optimization

LLM PLANNING ASSISTANT

Explain evidence
Compare scenarios
Triage validation

ML estimates. Deterministic models quantify. City teams control policy. The LLM explains and triages evidence.

The assistant never writes scores, rankings, climate values, weights, or portfolio membership.

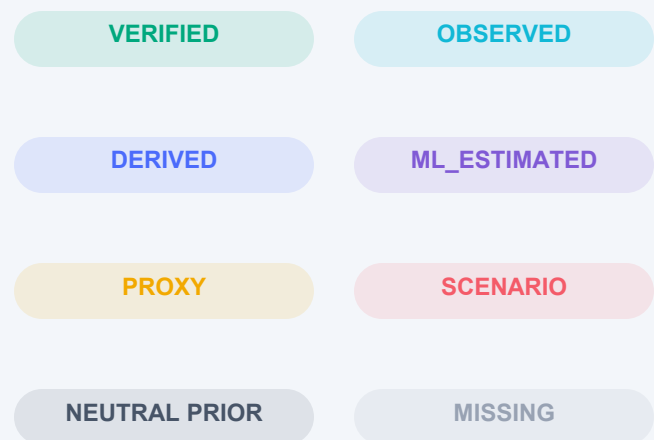
DATA, PROVENANCE, VALIDATION

Every recommendation carries an evidence trail.

7 PRIMARY SOURCE FAMILIES

- Historic Sakay GTFS
- WorldPop 2020
- DOE climate + feasibility
- OSM routes + infrastructure
- LTFRB planning sources
- Operator references + ledgers
- Validation + governance records

VISIBLE FIELD STATUS

**20**

dated OSM matches

22

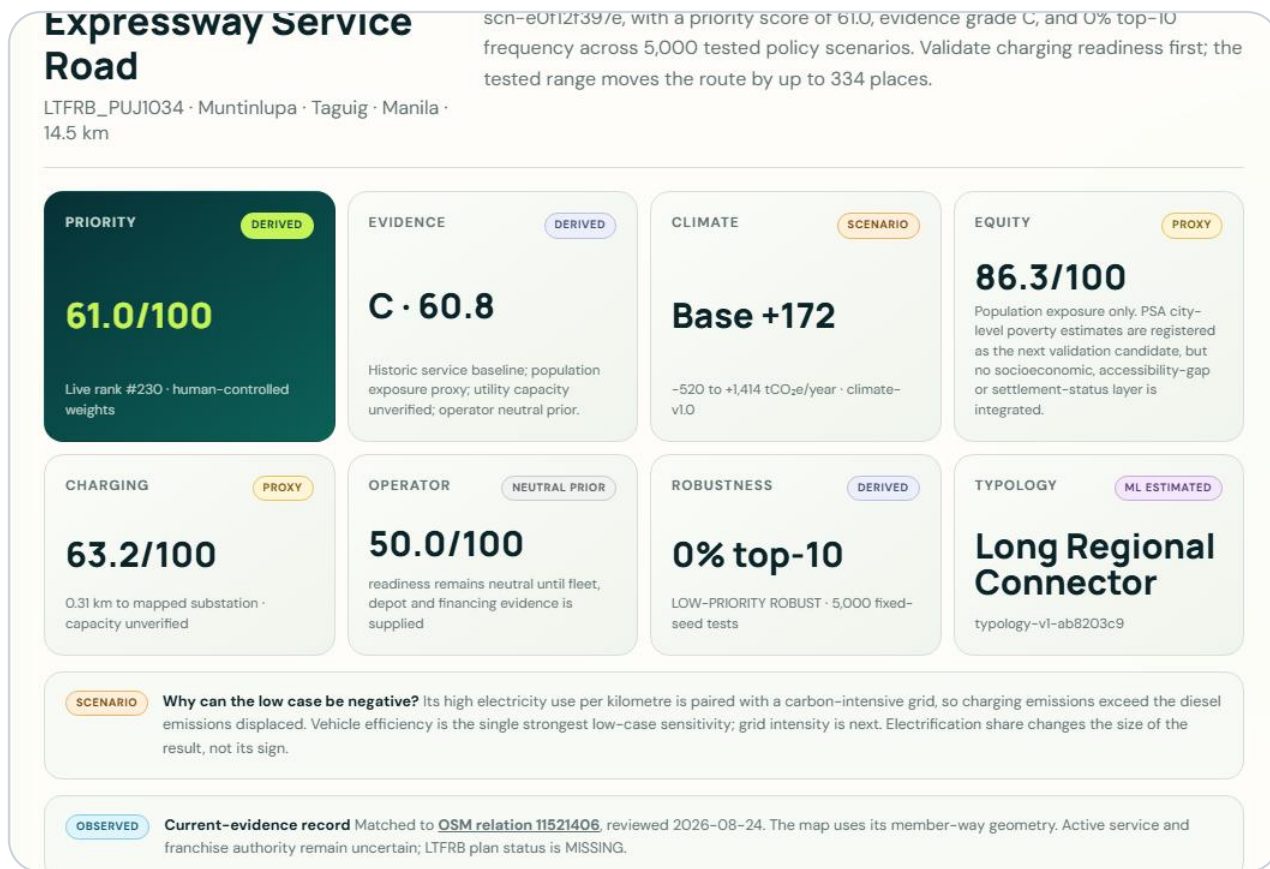
source geometries

18

registered source records

CURRENT-EVIDENCE VALIDATION LAYER

A dated external route record now changes the evidence state — without claiming active service.



HISTORIC ONLY

1,522-route screening baseline

↓ fuzzy endpoint review + one-to-one relation acceptance

OBSERVED

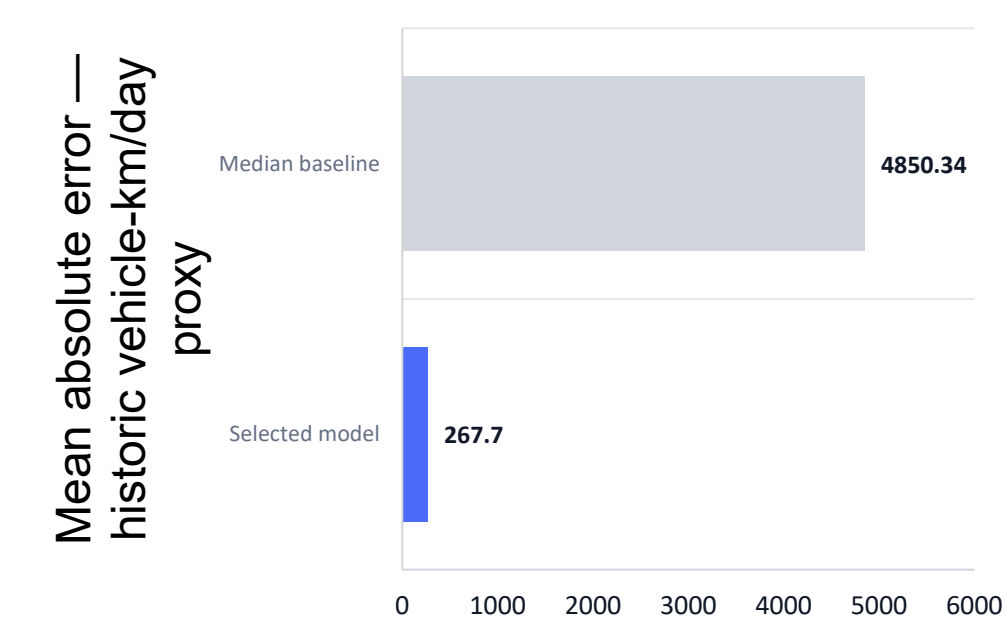
20 corridors with dated OSM route relations

route=share_taxi convention
 edited 2023-01-01 or later
 actual member-way geometry
 active service remains uncertain

Example: LTFRB_PUJ1034 · OSM relation 11521406 · reviewed 24 Aug 2026

ML SERVICE INTELLIGENCE

Machine learning estimates service intensity where evidence is incomplete.



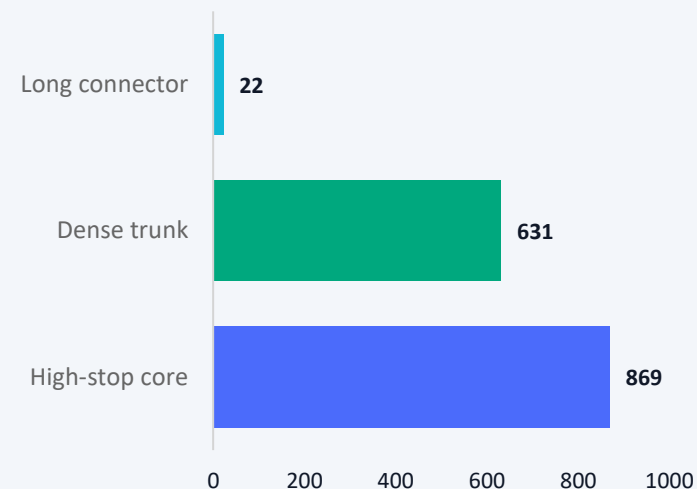
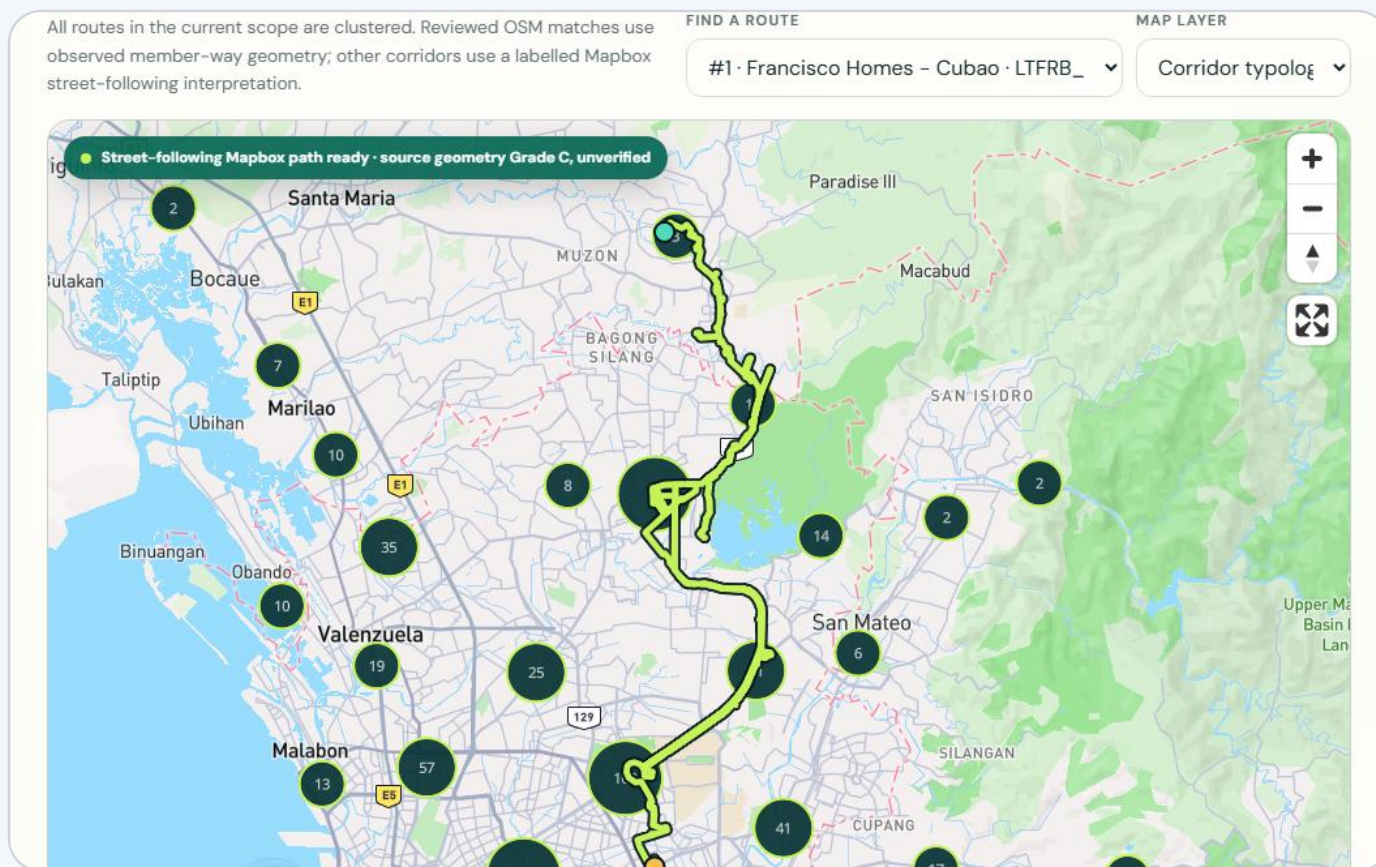
HistGradientBoostingRegressor	
TARGET	Historic schedule-based vehicle-km/day proxy
VALIDATION	GroupKFold(5) by normalized corridor
ROWS	1,521 training records · 714 corridor groups
METRICS	MAE 267.70 · RMSE 574.14 · R² 0.9907
VERSION	service-v1-266e0b1d

Only LTFRB_PUJ2451 uses ML: historic activity is MISSING; the model supplies 6,657 vehicle-km/day for the climate input. It is not headway, current service, or ridership.

Source: model_metrics.json · grouped corridor validation · no score/rank leakage features

CORRIDOR TYPOLOGY

Route2Zero compares like with like.



SELECTED K = 3

Silhouette 0.373

KMeans · typology-v1-ab8203c9
Typology informs comparison, not policy score.

CLIMATE + ENERGY IMPACT

The system estimates climate outcomes — not an ‘emissions score.’

+368

base tCO₂e/year



31,585

historic daily VKT



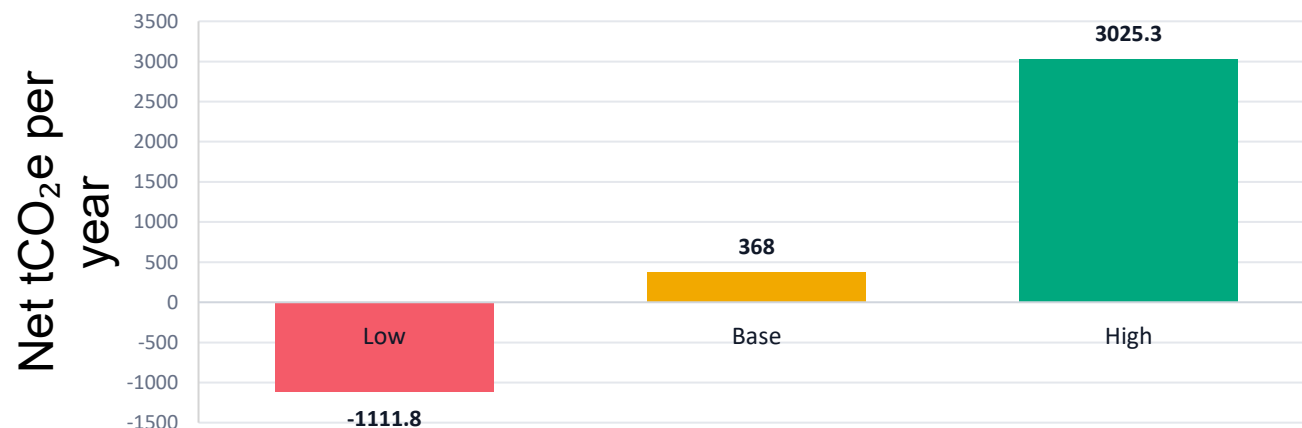
50%

base electric share



**-1,112 to
+3,025**

bounded range



BASE CASE FIRST · SCENARIO

The low case turns negative when a carbon-intensive grid is paired with an inefficient EV assumption. In that combination, electricity emissions exceed avoided diesel emissions.

Dominant sensitivity: vehicle efficiency, together with grid intensity.

EQUITY + ACCESSIBILITY

Climate value is not enough if access is left behind.

77.42

population-exposure score

25 / 100

equity evidence confidence

PROXY

claim status

What exists

WorldPop population exposure
Route-level spatial overlap
Transparent missing-data policy

What does not yet exist

Validated socioeconomic layer
Accessibility-gap or transit-dependence layer
Settlement-status evidence

Next candidate: PSA Small Area Poverty Estimates at city level. It is registered but MISSING from scoring until route-city geometry and aggregation rules are validated.

CHARGING + OPERATOR READINESS

A climate opportunity is not implementation-ready without infrastructure and operators.

CHARGING EVIDENCE

60.45

readiness score · PROXY

Energy need

13.16 MWh/day · base

Nearest mapped substation

0.56 km

Nearest mapped charger

0.65 km

Candidate terminals

2 · unverified

OPERATOR EVIDENCE

1 / 8

named desk reference

Named reference

PUJ1405

Consent-based readiness

0 / 8

Evidence confidence

5 / 100

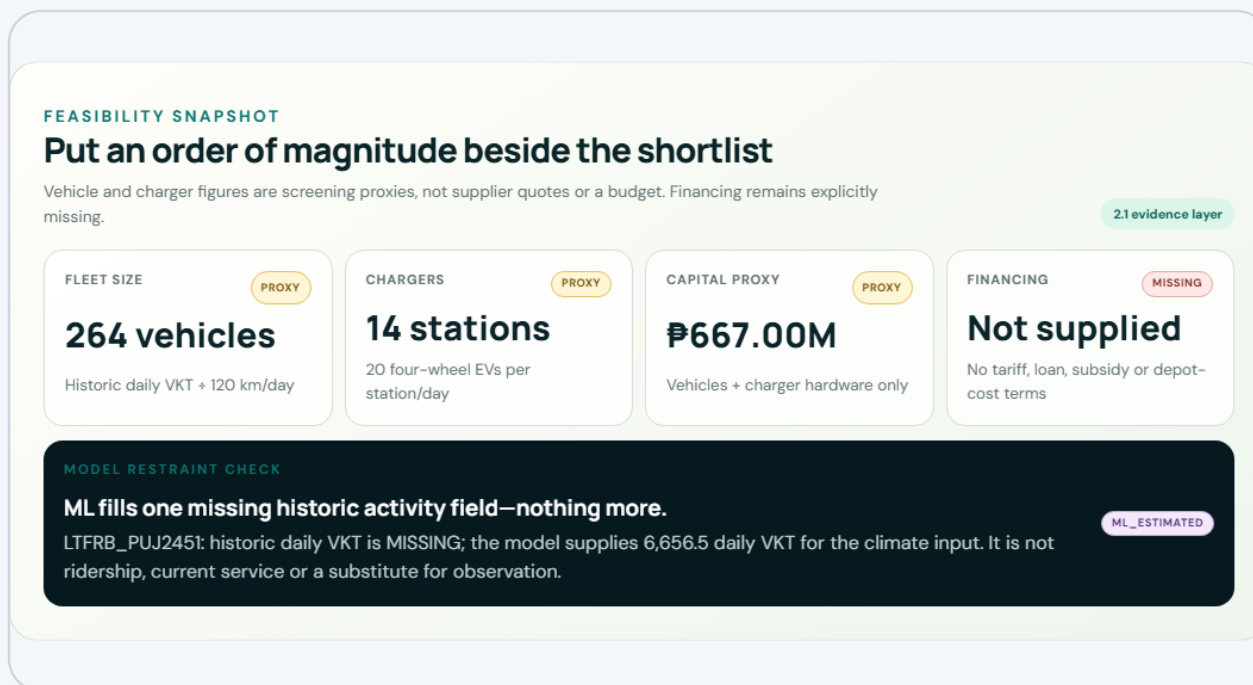
Readiness status

NEUTRAL PRIOR

UTILITY CAPACITY: NOT VERIFIED — mapped proximity is not capacity, site control, or interconnection approval.

FEASIBILITY + COST / FLEET SNAPSHOT

Put an order of magnitude beside the shortlist — without calling it a budget.



1,943

vehicles

102

chargers

₱4.91B

vehicle + charger proxy

MISSING

financing terms

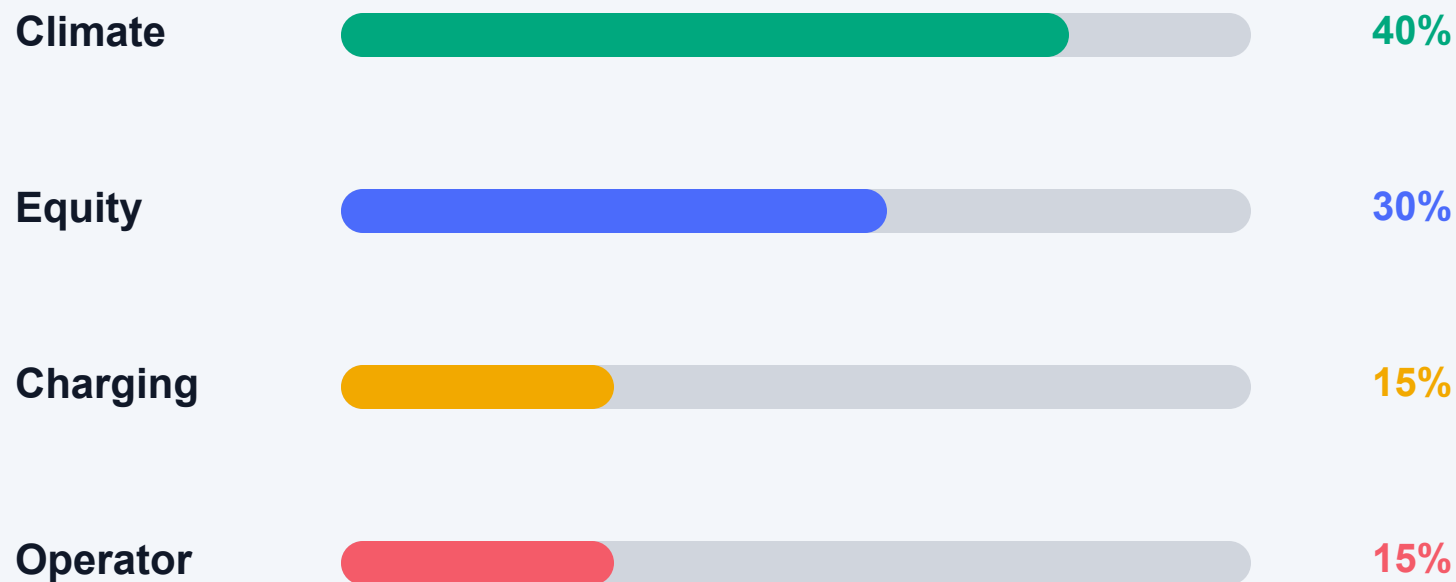
PROXY assumptions

120 km/vehicle-day · ₱2.5M/vehicle · 20 vehicles/charger-day · ₱0.5M/charger

Excluded: land/depot, civil works, grid upgrades, interconnection, battery replacement, taxes, insurance, financing, and O&M. Confirm fleet, duty cycle, tariff, depot, and financing before any decision.

HUMAN-CONTROLLED POLICY MODEL

The policy trade-offs stay visible.



DEFAULT SCENARIO

Climate + Equity

scn-e0f12f397e

Named alternatives

Equity-first: 25 / 50 / 15 / 10

Infrastructure-first: 25 / 20 / 35 / 20

Weights are normalized to 100%. They are a city choice — never presented as objectively correct.

UNCERTAINTY + RANK STABILITY

A recommendation is stronger when it survives changing assumptions.

5,000

fixed-seed policy-weight
simulations

Francisco Homes–Cubao

rank 1



Binangonan–JRC via Angono

rank 11



Rank stability measures resilience to tested policy weights — not model accuracy.

VALUE OF INFORMATION

Route2Zero tells the city which missing evidence is worth collecting first.

Climate assumptions

rank swing up to 1,395

Portfolio flip possible

Equity population exposure

rank swing up to 7

Validate area evidence

Charging readiness

rank swing up to 3

Request utility/site evidence

Operator readiness

rank swing up to 3

Interview operator/cooperative

Next action: calibrate vehicle efficiency, electrification share, and grid assumptions before treating the flagship climate case as decision-ready.

CONSTRAINED PHASE-1 PORTFOLIO

Cities pilot programs, not leaderboards.

SIMPLE TOP 8

PUJ1352 removed

PUJ1240 removed

PUJ2084 removed

PUJ1157 removed

CONSTRAINED 8

PUJ1638 added

PUJ1153 added

PUJ1350 added

PUJ1405 added



MAX 8

MAX 2 / CITY

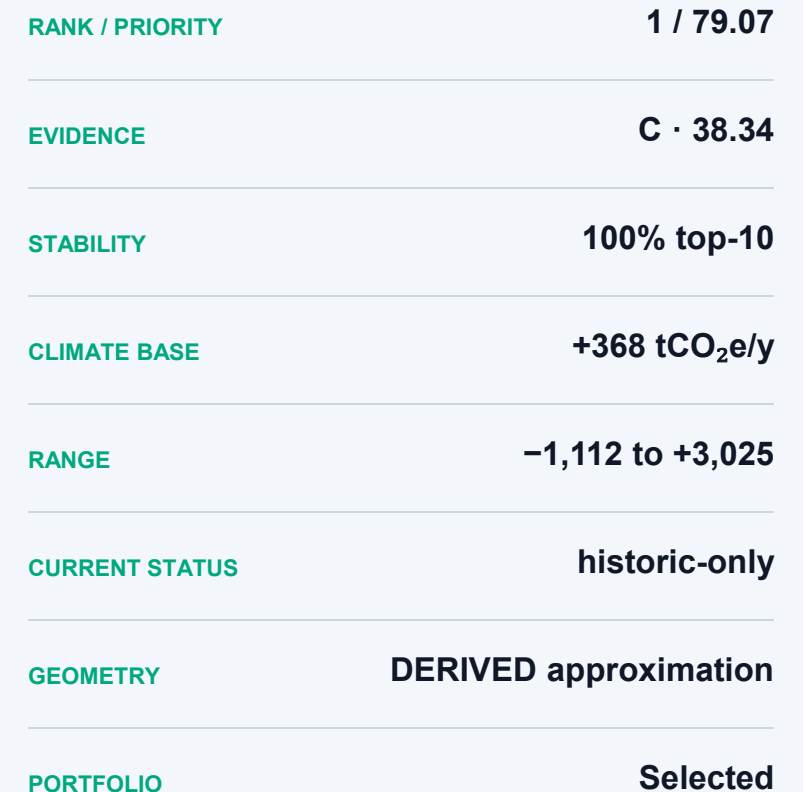
MAX 1 DIRECTION / CORRIDOR

EQUITY ≥ 40

EVIDENCE ≥ C

BASE CASE
 +2,711 tCO₂e/year

Bounded low–high: –8,191 to +22,288 · avg equity 73.9 · all 8 grade C



r2z-d4c8d4cc709a · scn-e0f12f397e · prt-fd6de9d793

PILOT · IMPACT · DIFFERENTIATION · SCALE

Six months to turn a defensible shortlist into city-owned evidence.



EXPECTED IMPACT

8 corridors enter deeper validation
Base +2,711 tCO₂e/year scenario
Evidence improvement is measured

WHY ROUTE2ZERO

Not one black-box score
Field-level status + uncertainty
Humans control policy

HOW IT SCALES

Metro Manila is the deep pilot
Local source adapters
Every city recalibrates

Team Larpers

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